

Assignment 4 SPM (Jan-April 2026)

Side-Channel Attacks

Kyber (ML-KEM)

Post-quantum cryptography aims to develop cryptographic schemes that remain secure against adversaries equipped with quantum computers, which can break many classical public-key systems such as RSA and ECC. Among the leading candidates is CRYSTALS-Kyber, a lattice-based key encapsulation mechanism selected for standardization due to its strong security guarantees and efficient performance yet vulnerable to Power Side Channels.

The traces are based on implementation of Kyber (ML-KEM) polynomial pointwise multiplication (PWM): $S_1 \cdot r$. The details about the implementation are provided below. Your task, should you choose to accept it, is to use Correlation Power Analysis (CPA) to recover the secret polynomial S_1 given ciphertext polynomials r and corresponding power traces.

1 Dataset Description

This section describes the datasets used for performing the Correlation Power Analysis (CPA) attack on the Kyber polynomial pointwise multiplication.

1.1 Ciphertext Polynomial Data: `polynomials.csv`

- This file contains the ciphertext polynomial coefficient values r used during the polynomial pointwise multiplication (PWM).
- Each row contains 256 coefficients representing the polynomial.
- All coefficients are in the Kyber modulus domain, i.e., modulo $q = 3329$.

1.2 Power Trace Data: `power_traces.csv`

- This file contains power traces corresponding to the polynomial pointwise multiplication in the Number Theoretic Transform (NTT) domain.
- Each row corresponds to the same execution index as in `polynomials.csv`.
- Each column represents a time sample of the power consumption.

Tasks

1. For the given dataset, retrieve all 256 coefficients of secret polynomial S_1 . We provide partial coefficients (i.e, 64) for validation (see hint section). (25 marks)
2. The implementation must complete the coefficient retrieval process within 300 seconds. (5 marks)
3. Your implementations are tested against hidden test case having noisy traces, so make your implementations robust enough. (20 marks)
4. Documentation of code/prompts and write up mentioning the strategy used to recover S_1 . (20 marks)

5. Answer the following questions.

- (a) Plot graphs showing how the correlation score of correct and wrong polynomial coefficients changes with the number of traces. The graph should have correlation score on y-axis and number of traces on x-axis. (10 marks)
- (b) Perform CPA using Hamming Weight leakage model on Montgomery multiplication intermediates (m_0, m_1, m_2, m_3). Compare performance across different intermediates. (10 marks)
- (c) Experiment with alternative leakage model (Hamming Distance) and compare key recovery success rates for Hamming weight and Hamming Distance using Guessing Entropy metric. (10 marks)

Submission Rules

- The submitted code must be automated in a way that it should take inputs in the given format and it should give a $\langle \text{rollnumber_key} \rangle$.txt file having all the coefficients recovered in a list format (i.e [1,2,...])
- Your python script will be run using `python3  $\langle \text{rollnumber} \rangle$ .py polynomials.csv power traces.csv`
- We will use the 'diff' command on your submissions, so make sure your .txt file is in the prescribed format.
- If you have used LLMs for this assignment: (a) mention the LLMs that you used, (b) document what you wanted LLM for, (c) the prompts that you used.
- Code and LLM prompts (if used) should be well documented, clean and appropriate comments should be added.

Input format

CSV files as provided in the dataset:

- *polynomials1_traces.csv*: Ciphertext polynomial r coefficients.
- *output_hw NTT_traces.csv*: Power traces.
- `python3  $\langle \text{rollnumber} \rangle$ .py polynomials.csv power traces.csv`

Output format

- $\langle \text{rollnumber_key} \rangle$.txt
- Complete 256 coefficients of secret polynomial S_1
Example: [1,14,356,76,34,.....543,23,257]

The contacts of TA are attached in the file and you are free to contact them for any doubts.
Remember: hardening of key or any unethical practices found will result in 0 marks.

Note: If you have used any LLM's you must include the exact prompts used in the write up.

Hint

You can use the `partial_secret_key.txt` file.

2 Submission Folder

The submission folder must contain 3 files:

- python implementation in the name format as mentioned.(.py)
- Documentation (.pdf)
- The coefficients file (.txt)

3 Implementation of Kyber

Please refer to the Kyber source code for the values of ω used throughout Kyber PWM (<https://github.com/pq-crystals/kyber>).

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